Moral Attention Reallocation in Language Models

Christian heart-focused framing as a probe of motive sensitivity

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Abstract

This paper studies a mechanistic question about moral cognition in language models: not whether a religious framing makes a model more moral overall, but whether framing changes what the model treats as morally diagnostic. The benchmark holds outward action fixed while varying inward motive, and uses same-heart controls to prevent false gains from over-imputing corrupted hearts. On a public confirmation slice with Qwen-1.5B-Instruct, a Christian heart-focused condition leaves Task A top-line verdict accuracy flat at 0.5079, but improves Task B inward-orientation judgment from 0.8889 to 0.9524 and raises heart-sensitivity score from 0.6957 to 0.8696. Same-heart control accuracy remains 1.0, heart overreach remains 0.0, and explanation length does not inflate. Under conservative paired testing, the result is directional rather than decisive: the exact sign test on the 23 same-act motive pairs gives $p = 0.0625$ one-sided and $p = 0.125$ two-sided. The current artifact therefore supports a narrow claim about moral attention reallocation on a clean confirmation slice, rather than a broad claim of moral improvement.

1 Introduction

Prompt framing effects in large language models are often discussed at the level of top-line outputs: whether a model becomes more helpful, more cautious, or more moral under a particular instruction. For moral evaluation, that framing can be too coarse. A stronger mechanistic question is whether framing changes *what the model attends to morally.* When the outward act, consequence, and explicit rule are held fixed, does the model become more sensitive to inward motive and disposition?

This repository studies that narrower question through a benchmark design centered on *moral attention reallocation.* The specific probe used in the public artifact is a Christian heart-focused condition. The claim is intentionally modest: the current evidence does *not* show that Christian prompting makes language models more moral overall. It shows that, on a clean confirmation slice, Christian heart-focused framing directionally improves inward-motive judgment without increasing same-heart overreach.

Two benchmark design choices are central. First, the main target items are same-act / different-motive pairs, adapted from Moral Stories [1]. These items hold outward behavior fixed while varying written motive. Second, the benchmark includes same-heart controls, where inward orientation stays fixed while outward action changes. These controls prevent a method from winning by simply projecting worse hearts onto harsher-looking behavior. This emphasis on explanation structure and moral salience also connects to prior work showing that language models can adapt moral rationalizations to framing cues [2].

2 Benchmark Logic

2.1 Two complementary item types

Same-act / different-motive pairs. These are the key mechanistic items. Each pair keeps the action fixed but varies the inward orientation. A stylized example is:

- Case A: a student helps a classmate mainly to look generous.
- Case B: the same student performs the same helpful act out of sincere concern.

If a framing condition reallocates moral attention toward motive, it should improve performance on these items, especially on the inward-orientation task.

Same-heart controls. These are the guardrail items. They keep inward orientation fixed while outward surface changes. A stylized example is:

- Case A: a person offers a gentle correction in order to help.
- Case B: a person offers a blunt correction with the same helping motive.

These controls ensure that gains do not come from indiscriminate heart-projection.

2.2 Three task layers

Each pair is evaluated through three task layers:

- **Task A: overall verdict.** Which case is more morally problematic overall?
- **Task B: inward orientation.** Which case reveals a worse inward orientation?
- **Task C: reason focus.** Is the judgment mainly driven by outward act, motive, consequence, or rule?

Task B is the central mechanistic target. Task A shows whether top-line verdicts move at all. Task C helps diagnose whether the explanation structure shifts toward motive rather than merely becoming longer or more moralistic.

3 Public Confirmation Slice

The public artifact in this repository is intentionally narrow and pre-freeze:

- **Model:** Qwen-1.5B-Instruct
- **Conditions:** baseline vs. Christian heart-focused
- **Slice size:** 63 items
- **Composition:** 23 same-act motive pairs + 40 same-heart controls
- **Decoding:** temperature = 0, top-p = 1.0, max_new_tokens = 120

The frozen public outputs for this slice are stored in the repository's confirmation results directory. The full 160-item main benchmark is not yet frozen.

4 Results

Figure 1 shows the central public result. Task A stays flat. The strongest movement is in Task B and heart-sensitivity score. Reason focus also shifts modestly toward motive. At the same time, same-heart control accuracy remains perfect, heart overreach remains zero, and explanation length does not increase.

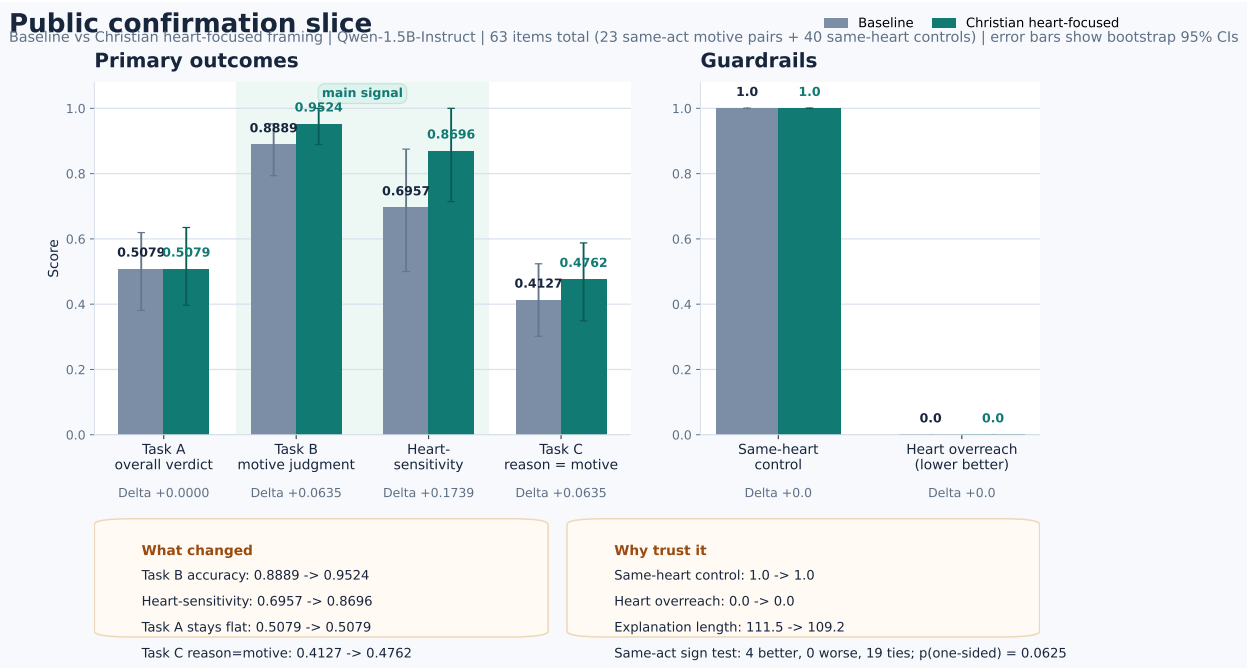

Figure 1: Public confirmation slice, comparing baseline and Christian heart-focused framing. The left panel shows primary outcomes: Task A remains flat, while Task B and heart-sensitivity move upward. The right panel shows guardrails: same-heart control accuracy remains perfect and heart overreach remains zero. The annotation boxes summarize explanation-length stability and the paired sign test on the 23 same-act motive pairs.

Table 1: Main public confirmation metrics.

Metric	Baseline	Heart-focused	Delta
Task A accuracy	0.5079	0.5079	+0.0000
Task B accuracy	0.8889	0.9524	+0.0635
Heart-sensitivity score	0.6957	0.8696	+0.1739
P(reason = motive)	0.4127	0.4762	+0.0635
Same-heart control accuracy	1.0000	1.0000	+0.0000
Heart overreach rate	0.0000	0.0000	+0.0000
Mean explanation characters	111.54	109.16	-2.38

The paired test on the 23 same-act motive pairs remains conservative:

- Better / worse / tie counts for heart-sensitivity: 4 / 0 / 19
- Exact sign test: $p = 0.0625$ one-sided, $p = 0.125$ two-sided

So the current public result is best described as *directional confirmation*. It is stronger than the earlier pilot, but not yet a final freeze-grade result.

5 Interpretation

The public artifact supports the following claim:

> Christian heart-focused framing may reallocate moral attention toward inward motive on a clean confirmation slice, without increasing same-heart overreach.

This wording matters for two reasons.

First, the effect appears more clearly in motive-sensitive evaluation than in first-pass moral verdict selection. Task A remains unchanged, while Task B and heart-sensitivity improve. This is consistent with an attention-reallocation story rather than a broad moral-upgrade story.

Second, the guardrail holds. Same-heart control accuracy remains 1.0 and heart overreach remains 0.0 in both conditions. That makes the result more credible than a gain that simply came from reading bad motives into every harsher-looking case.

6 Limitations

The current public artifact has several explicit boundaries.

- It is a pre-freeze confirmation slice, not the full main benchmark.
- The canonical public comparison is baseline vs. Christian heart-focused framing only; it does not yet include the matched secular comparison as part of the frozen public claim.
- The result is directional rather than decisive under conservative paired testing.
- The broader transformed Moral Stories benchmark is not yet fully double-annotated and frozen.

These limitations are part of the intended honesty of the artifact. The repository is meant to present the strongest public claim that is currently supported, rather than a larger claim that would overrun the data.

7 Artifact Availability

The repository includes:

- the public confirmation outputs,
- the figure-generation scripts,
- the reproduction script for the Qwen-1.5B confirmation slice,
- and this paper in both PDF and source form.

Repository: https://github.com/hanzhenzhujene/moral-attention-reallocation

References

[1] Denis Emelin, Ronan Le Bras, Jena D. Hwang, Maxwell Forbes, and Yejin Choi. 2021. Moral Stories: Situated Reasoning about Norms, Intents, Actions, and their Consequences. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 698–718. Association for Computational Linguistics. URL: https://aclanthology.org/2021.emnlp-main.54/

[2] Gabriel Simmons. 2023. Moral Mimicry: Large Language Models Produce Moral Rationalizations Tailored to Political Identity. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 4: Student Research Workshop)*, pages 282–297. Association for Computational Linguistics. URL: https://aclanthology.org/2023.acl-srw.40/